Title: UK Greenhouse Gas Emission Trends (1970-2023)

BACKGROUND: This analysis explores historical UK GHG emissions across 9 sectors. While overall emissions are dropping, the goal of this study is to identify which specific sectors are driving the success and which are lagging, posing a threat to future climate targets.

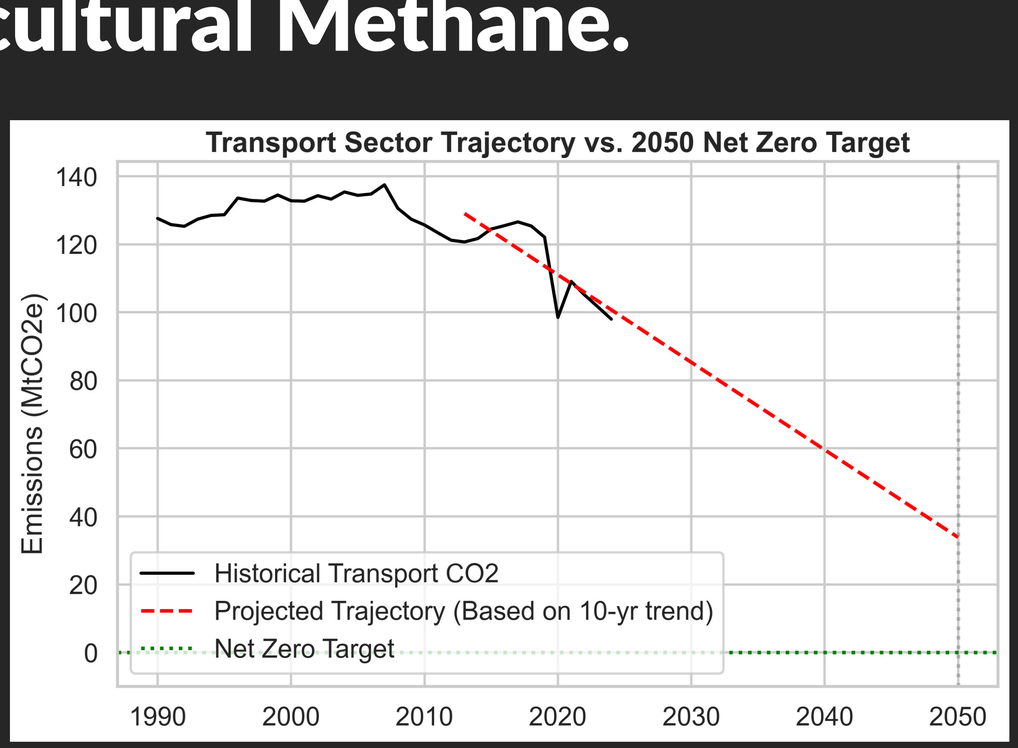
METHODS

1. **Python EDA:** Cleaned and aggregated historical data across 9 sectors.
2. **Rolling 5-Year CAGR:** Calculated to measure the acceleration/deceleration of emission drops.
3. **Linear Regression:** Applied to Transport data to forecast the 2050 Net Zero trajectory.

RESULTS

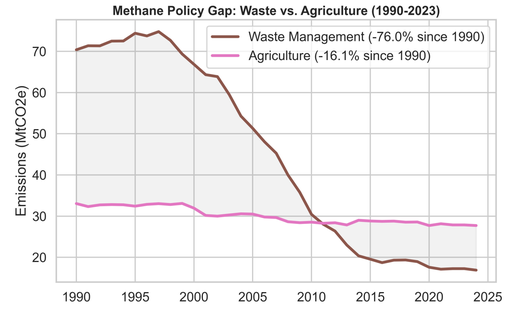
- **Energy Success:** The Energy Supply sector drove the largest absolute reductions, dropping at the fastest rate (CAGR) since 1990.
- **The Transport Shock:** Transport CO2 saw a massive lockdown drop in 2020 but rebounded sharply, making it the #1 overall CO2 emitter today.
- **The Methane Gap:** While Waste Management successfully slashed methane emissions, Agricultural methane has completely flatlined over the last 30 years.
- **Trajectory Warning:** Linear forecasting reveals that Transport will miss the 2050 Net Zero legal target on its current 10-year trend.

While the UK rapidly decarbonized Energy Supply, the 2050 Net Zero target is jeopardized by stubborn emissions in Road Transport and Agricultural Methane.

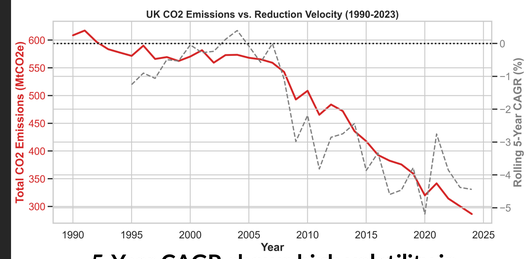


Spoiler: At its current 10-year trajectory, the Transport sector will fail to reach zero emissions, completely missing the UK's 2050 Net Zero mandate.

AMMO BAR



Waste Management slashed methane, but Agricultural methane has flatlined.



5-Year CAGR shows high volatility in emission reduction rates.

Recommendation for In-Depth Study:
The forecasted failure of the Transport sector and the 30-year stagnation of Agricultural methane require deep-dive investigations into sector-specific policy barriers, as current broad-stroke interventions are failing here.

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